Week #: Week 9

Subject: DP Chemistry Unit Title: Reactivity 3 Grade: 12

Lessons	Learning targets	Assessment tools and strategies Formative/ Summative/ Differentiation	Resources and links for Parents
Lesson 1: 3.1.7-3.1.8 Lab	Student apply their knowledge in the form of a lab report to deduce the neutralisation reaction as well as PH graphing.	Formative : Lab report	
Lesson 2: 3.1.7-3.1.8	• Formulate equations for the reactions between acids and metal oxides, metal hydroxides, hydrogen carbonates and carbonates. • Sketch and interpret the general shape of the pH curve.	Student task Peer Assessment; Exit Ticket, Revision Village Practice questions on topic.	IB Chemistry: InThinking/ Revision Village/ PPT, One note(Teams) in class notebook under Week 9_L2
Lesson 3: 3.2.3-3.2.4	Describe and carry out neutralization reactions between acids and bases, write balanced word and symbol equations, and explain how these reactions are used in everyday life and in the UAE.	Student task Peer Assessment; Exit Ticket, Revision Village Practice questions on topic.	IB Chemistry: InThinking/ Revision Village/ PPT, One note(Teams) in class notebook under Week 9_L3
Lesson 4: 3.2.5	• Deduce oxidation states of an atom in a compound or an ion. • Identify the oxidized and reduced species and the oxidizing and reducing agents in a chemical reaction.	Student task Peer Assessment; Exit Ticket, Revision Village Practice questions on topic.	IB Chemistry: InThinking/ Revision Village/ PPT, One note(Teams) in class notebook under Week 9_L4

Lesson 5:HL 3.1.9-3.11	• Interconvert [H+], [OH–], pH and pOH values. • Interpret the relative strengths of acids and bases from these data. • the relationship ***K*a** × ***K*b = *K*w** can be derived • from the expressions for *K*a and *K*b.	Student task Peer Assessment; Exit Ticket, Revision Village Practice questions on topic.	IB Chemistry: InThinking/ Revision Village/ PPT, One note(Teams) in class notebook HL under Week 9_L1
Homework	Revise past papers on short data based questions on Inthinking or revision village.		
Teacher's notes/ Reminders	**Please ensure student has a copybook, calculator and stationary.**		

Technology Integration	PPT,Hand on activity
Cross-curricular links	**Physics:** electron flow and potential difference **Environmental Science:** pH control in water systems **Mathematics:** logarithmic and proportional relationships
External Assessment Integration	**Preparation for DP Paper 2 (structured questions on titration, buffers, and redox) and Paper 3 (data analysis involving cell potentials).**

Wellbeing/ ATL Skills	**Thinking:** applying models to abstract chemical systems **Research:** investigating UAE applications of sustainable chemistry **Self-Management:** accurate data handling and lab discipline
Standards	**DP Chemistry Reactivity 3**
Inquiry/ TOK Question	*How does our understanding of redox and acid–base reactions influence innovation in sustainable energy technologies?*
Success Criteria (Measurable)	I Can: • **solve** pH, Ka, and E°cell problems. • **explain** reaction processes using correct IB terminology. • **evaluate** the role of chemistry in UAE sustainability.
National Identity Domain	**Citizenship:** Apply knowledge of redox and acid–base systems to understand the UAE's renewable energy and desalination initiatives. **Values:** Promote responsibility and innovation through the study of sustainable technologies like solar batteries and pH control in agriculture. **Culture:** Recognize the UAE's heritage of adaptation and technological advancement, linking traditional water management systems to modern chemical applications.